Exp 2 - Design and Performance Analysis of Chebyshev Type-I IIR Low-Pass and High-Pass Filters Using MATLAB

Program :

```
clc;
clear;
close all;

%% ============================================================
% CHEBYSHEV TYPE-I IIR FILTER DESIGN
% LOW-PASS AND HIGH-PASS WITH BUTTERWORTH COMPARISON
% ============================================================

%% Input Specifications

Fs = 1000;          % Sampling frequency (Hz)

Fp = 100;           % Passband edge (Hz)
Fst = 150;          % Stopband edge (Hz)

Rp = 1;             % Passband ripple (dB)
Rs = 40;            % Stopband attenuation (dB)


%% ============================================================
% NORMALIZED FREQUENCIES
% ============================================================

Wp = Fp/(Fs/2);
Ws = Fst/(Fs/2);


%% ============================================================
% CHEBYSHEV TYPE-I LOW-PASS FILTER
% ============================================================

[N_C_LP, Wn_C_LP] = cheb1ord(Wp, Ws, Rp, Rs);

[b_C_LP, a_C_LP] = cheby1(N_C_LP, Rp, Wn_C_LP, 'low');


%% ============================================================
% CHEBYSHEV TYPE-I HIGH-PASS FILTER
% ============================================================

% For high-pass, passband must be greater than stopband
Wp_HP = Fst/(Fs/2);
Ws_HP = Fp/(Fs/2);

[N_C_HP, Wn_C_HP] = cheb1ord(Wp_HP, Ws_HP, Rp, Rs);

[b_C_HP, a_C_HP] = cheby1(N_C_HP, Rp, Wn_C_HP, 'high');


%% ============================================================
% BUTTERWORTH LOW-PASS FOR COMPARISON
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% ========================================================

[N_B_LP, Wn_B_LP] = buttord(Wp, Ws, Rp, Rs);

[b_B_LP, a_B_LP] = butter(N_B_LP, Wn_B_LP, 'low');


%% ========================================================
% BUTTERWORTH HIGH-PASS FOR COMPARISON
% ========================================================

[N_B_HP, Wn_B_HP] = buttord(Wp_HP, Ws_HP, Rp, Rs);

[b_B_HP, a_B_HP] = butter(N_B_HP, Wn_B_HP, 'high');


%% ========================================================
% DISPLAY FILTER ORDERS AND COEFFICIENTS
% ========================================================

fprintf('\n============================================\n');
fprintf(' CHEBYSHEV TYPE-I IIR FILTER DESIGN\n');
fprintf('============================================\n');

fprintf('\nCHEBYSHEV TYPE-I LOW-PASS FILTER\n');
fprintf('Filter Order = %d\n', N_C_LP);
fprintf('Normalized Cutoff Frequency = %.4f\n', Wn_C_LP);

fprintf('\nNumerator Coefficients (b):\n');
disp(b_C_LP);

fprintf('Denominator Coefficients (a):\n');
disp(a_C_LP);


fprintf('\nCHEBYSHEV TYPE-I HIGH-PASS FILTER\n');
fprintf('Filter Order = %d\n', N_C_HP);
fprintf('Normalized Cutoff Frequency = %.4f\n', Wn_C_HP);

fprintf('\nNumerator Coefficients (b):\n');
disp(b_C_HP);

fprintf('Denominator Coefficients (a):\n');
disp(a_C_HP);


fprintf('\n============================================\n');
fprintf(' BUTTERWORTH COMPARISON\n');
fprintf('============================================\n');

fprintf('\nButterworth Low-Pass Order  = %d\n', N_B_LP);
fprintf('Butterworth High-Pass Order = %d\n', N_B_HP);


%% ========================================================
% FREQUENCY RESPONSES
% ========================================================

[H_C_LP, f] = freqz(b_C_LP, a_C_LP, 2048, Fs);
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[H_C_HP, ~] = freqz(b_C_HP, a_C_HP, 2048, Fs);

[H_B_LP, ~] = freqz(b_B_LP, a_B_LP, 2048, Fs);
[H_B_HP, ~] = freqz(b_B_HP, a_B_HP, 2048, Fs);


%% ============================================================
% SINGLE OUTPUT WINDOW
% ============================================================

figure('Name','Chebyshev Type-I IIR Filter Analysis',...
       'NumberTitle','off');

%% 1. Chebyshev LPF Magnitude

subplot(3,2,1);

plot(f,20*log10(abs(H_C_LP)),'LineWidth',1.2);

grid on;

xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');

title('Chebyshev Type-I LPF - Magnitude');

xline(Fp,'--r','Passband');
xline(Fst,'--k','Stopband');


%% 2. Chebyshev HPF Magnitude

subplot(3,2,2);

plot(f,20*log10(abs(H_C_HP)),'LineWidth',1.2);

grid on;

xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');

title('Chebyshev Type-I HPF - Magnitude');

xline(Fp,'--r','Passband');
xline(Fst,'--k','Stopband');


%% 3. Chebyshev LPF Phase

subplot(3,2,3);

plot(f,unwrap(angle(H_C_LP)),'LineWidth',1.2);

grid on;

xlabel('Frequency (Hz)');
ylabel('Phase (rad)');

title('Chebyshev Type-I LPF - Phase');
```

```
%% 4. Chebyshev HPF Phase

subplot(3,2,4);

plot(f,unwrap(angle(H_C_HP)),'LineWidth',1.2);

grid on;

xlabel('Frequency (Hz)');
ylabel('Phase (rad)');

title('Chebyshev Type-I HPF - Phase');


%% 5. LPF Comparison

subplot(3,2,5);

plot(f,20*log10(abs(H_C_LP)),'LineWidth',1.2);
hold on;

plot(f,20*log10(abs(H_B_LP)),'--','LineWidth',1.2);

grid on;

xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');

title('LPF: Chebyshev vs Butterworth');

legend('Chebyshev Type-I','Butterworth');

xline(Fp,'--k');
xline(Fst,'--k');

hold off;


%% 6. HPF Comparison

subplot(3,2,6);

plot(f,20*log10(abs(H_C_HP)),'LineWidth',1.2);
hold on;

plot(f,20*log10(abs(H_B_HP)),'--','LineWidth',1.2);

grid on;

xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');

title('HPF: Chebyshev vs Butterworth');

legend('Chebyshev Type-I','Butterworth');

xline(Fp,'--k');
```

```
xline(Fst,'--k');

hold off;


%% ============================================================
% STABILITY ANALYSIS
% ============================================================

poles_C_LP = roots(a_C_LP);
poles_C_HP = roots(a_C_HP);

fprintf('\n============================================\n');
fprintf(' STABILITY ANALYSIS\n');
fprintf('============================================\n');

if all(abs(poles_C_LP) < 1)
    fprintf('Chebyshev LPF: STABLE\n');
else
    fprintf('Chebyshev LPF: UNSTABLE\n');
end

if all(abs(poles_C_HP) < 1)
    fprintf('Chebyshev HPF: STABLE\n');
else
    fprintf('Chebyshev HPF: UNSTABLE\n');
end
```

Output:

